Design of a Class D AM Transmitter at $1\,\mathrm{MHz}$ on a Single $6\,\mathrm{V}$ Supply

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***Abstract*—This paper presents the design and simulation of a Class D amplitude-modulation transmitter operating at $1\,\mathrm{MHz}$ on a single $6\,\mathrm{V}$ supply. The final stage dissipates $84\,\mathrm{mW}$ DC while delivering $13.0\,\mathrm{mW}$ into a $500\,\Omega$ load. Supply-rail modulation produces a $71\,\%$ modulation depth (FFT) at $1\,\mathrm{kHz}$ with $3.4\,\%$ total harmonic distortion. The $-3\,\mathrm{dB}$ audio passband covers $250\,\mathrm{Hz}$ to $5000\,\mathrm{Hz}$, and out-of-band emissions are more than $28\,\mathrm{dB}$ below the unmodulated carrier, exceeding the FCC Part 15 mask by $8\,\mathrm{dB}$. The transmitter uses six bipolar transistors and one LM358 op-amp, within the project's part budget.**

I. Introduction

Although amplitude modulation has been largely supplanted in modern communication, its educational value lies in combining nonlinear switching, RF tuned networks, and audio signal processing into a compact system. A Class D output stage suits the $100\,\mathrm{mW}$ DC budget because the switching transistors ideally dissipate no power. We followed a back-to-front design flow—tank Q and load first, then switches, RF buffer, and audio chain—and the paper follows the same order.

II. Architecture

Fig. 1 shows the complete transmitter schematic, comprising four functional blocks. An audio bandpass filter drives a complementary push-pull buffer through an LM358 unity-gain stage. The buffer feeds the audio transformer; its secondary, in series with an RFC between V_{DD} and the Class D supply rail, produces a low-frequency swing on the otherwise 6 V rail. An RF buffer (powered *from the modulated rail*, motivation in Section VII) converts the $1\,\mathrm{V_{pk}}$ carrier into base drive for the Class D switching pair, whose output passes through a series LC tank to the $500\,\Omega$ load.

III. Class D Output Stage

A. Series LC Tank

The tank's quality factor Q is set by the FCC Part 15 emission mask, which requires $\geq 20\,\mathrm{dB}$ attenuation at the band edges $510\,\mathrm{kHz}$ and $1707\,\mathrm{kHz}$. For a series-tuned RLC bandpass,

$$\left|\frac{H(f)}{H(f_0)}\right| = \frac{1}{\sqrt{1+Q^2\left(\dfrac{f}{f_0}-\dfrac{f_0}{f}\right)^2}}. \tag{1}$$

Evaluating at $1707\,\mathrm{kHz}$ gives $|f/f_0 - f_0/f| = 1.12$, and $510\,\mathrm{kHz}$ gives 1.45. The upper edge is therefore the limiting case; requiring $20\,\mathrm{dB}$ attenuation yields $Q \geq 9$. We chose $Q = 10$ to provide tolerance margin without compressing the audio sidebands at $\pm 5\,\mathrm{kHz}$ around the carrier.

With $R_L = 500\,\Omega$ and $\omega_0 = 2\pi \times 1\,\mathrm{MHz}$,

$$L_1 = \frac{QR_L}{\omega_0} = 796\,\mu\mathrm{H} \;\rightarrow\; 820\,\mu\mathrm{H}\,(\mathrm{standard}), \tag{2}$$

$$C_1 = \frac{1}{\omega_0^2 L_1} = 30.9\,\mathrm{pF} \;\rightarrow\; 30\,\mathrm{pF}\,(\mathrm{standard}), \tag{3}$$

giving a measured $Q = 10.3$ and $f_0 = 1.015\,\mathrm{MHz}$. The residual offset falls well inside the audio passband and is absorbed by the variable trimmer ($10\,\mathrm{pF}$ to $500\,\mathrm{pF}$).

B. Complementary Switching Pair

The output transistors are a 2N2907A high-side PNP (Q_1, emitter on rail) and a 2N2222A low-side NPN (Q_2, emitter on ground), driven 180°out of phase by the RF buffer. Base resistors $R_2 = R_3 = 2.2\,\mathrm{k\Omega}$ set quiescent base currents of approximately $200\,\mu\mathrm{A}$, sufficient to saturate each device given $\beta_{DC} > 100$ at peak collector currents on the order of $10\,\mathrm{mA}$.

The choice of $R_{2,3}$ has a strong effect on DC power: increasing from $1\,\mathrm{k\Omega}$ to $2.2\,\mathrm{k\Omega}$ dropped dissipation from $108\,\mathrm{mW}$ to $84\,\mathrm{mW}$, since lighter base loading allows fuller rail swing and reduces conduction overlap.

Speed-up capacitors C_9 (across R_2) and C_2 (across R_3), 330 pF each, accelerate switching transitions by bypassing the base resistor at high frequencies. Without them, the RC time constant formed by the base resistor and transistor input capacitance ($\sim$ 25 pF) gives $\tau \approx 55$ ns, which is significant against the $500\,\mathrm{ns}$ carrier half-period. This slow slew rate allows the switching node to traverse its active region gradually, causing brief shoot-through pulses that pull V_{rail} down through parasitic coupling, wasting DC power. At $1\,\mathrm{MHz}$, the capacitor reactance ($\approx 482\ \Omega$) falls well below $2.2\,\mathrm{k\Omega}$, so the fast edge bypasses the resistor and charges the base directly, while the resistor still sets the DC bias current limit. We found empirically that $680\,\mathrm{pF}$ over-couples and drives the transistor into deep saturation, causing excessive trough overshoot at turn-off, while $330\,\mathrm{pF}$ provides the optimal balance between edge speed and overshoot suppression.

IV. RF Input Buffer

The $1\,\mathrm{V_{pk}}$ RF source drives a 2N2222A common-emitter amplifier Q_3 through a $100\,\mathrm{nF}$ DC-blocking capacitor. The bias divider $R_7 = 7.5\,\mathrm{k\Omega}$ and $R_5 = 680\,\Omega$ is referenced to the modulated rail, placing Q_3 just below conduction threshold so it operates in Class C with $\sim 30\,\%$ duty cycle. This is

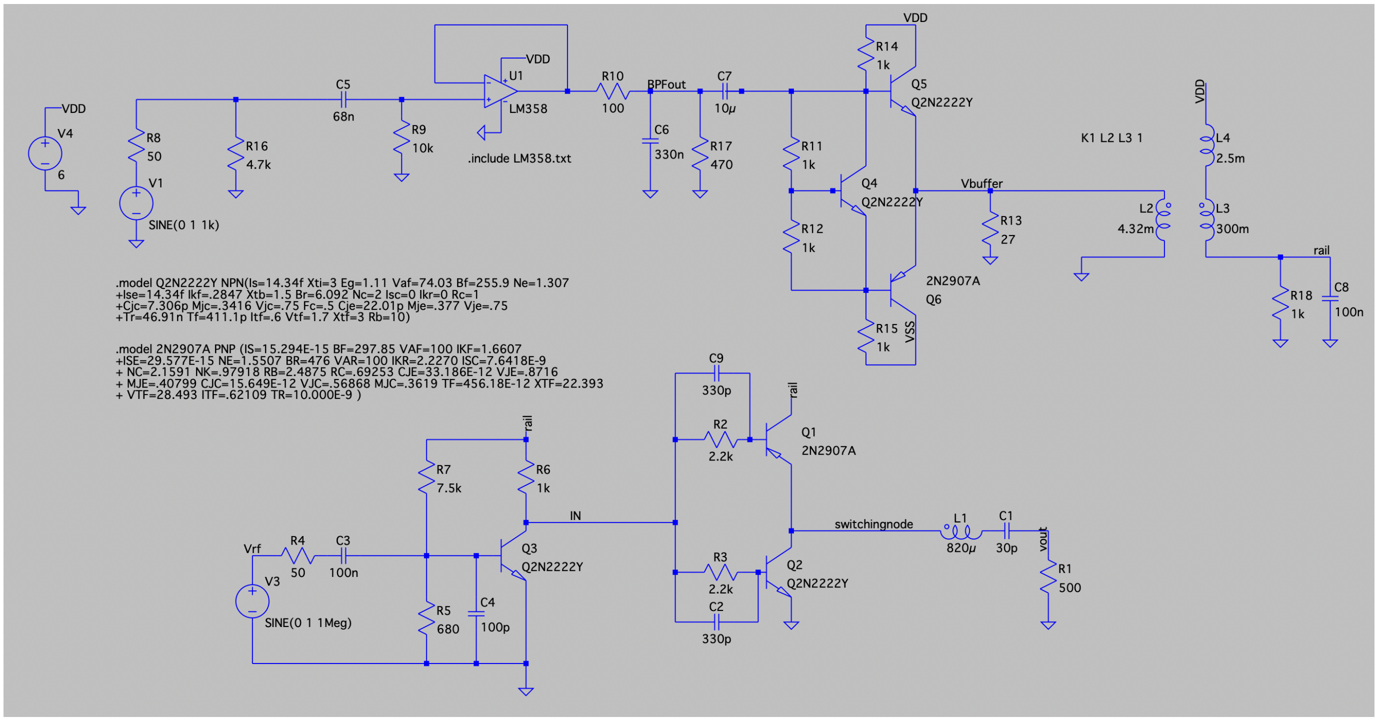

Fig. 1. Complete LTspice schematic. Top-left: audio bandpass filter and unity buffer (LM358). Top-center: complementary push-pull current buffer with V_{BE}-multiplier biasing (Q_4). Top-right: audio transformer (used in reverse for $8.33\times$ step-up) and rail network (RFC L_4, bypass C_8). Bottom-left: RF input buffer (Q_3, Class C). Bottom-right: Class D switching pair (Q_1, Q_2) and series-LC output tank (L_1, C_1) into $R_L = 500\,\Omega$.

acceptable because the high-Q tank rejects all harmonics, leaving only the $1\,\text{MHz}$ fundamental at the switching node.

A $100\,\text{pF}$ bypass capacitor C_4 across R_5 stabilizes the base AC reference. Because R_7 is tied to the modulated rail, residual switching ripple and audio-frequency variation would otherwise reach the base and jitter the Class C conduction threshold cycle-to-cycle. C_4 shunts these components to ground, locking the threshold and producing a consistent near-square waveform at the collector.

The collector resistor $R_6 = 1\,\text{k}\Omega$ delivers a rail-to-rail drive to the bases of Q_1 and Q_2. Because V_{R_5,R_7} scales with the rail, the buffer's quiescent point tracks the modulation and the carrier amplitude at the switching node remains a linear function of rail voltage; this linearity is the principal contributor to the low achieved THD.

V. Audio Modulation Chain

A. Bandpass Filter

The audio source provides $1\,\text{V}_{\text{pk}}$ through a $50\,\Omega$ series resistance. A passive high-pass section formed by $C_5 = 68\,\text{nF}$ and $R_9 = 10\,\text{k}\Omega$ sets the lower $-3\,\text{dB}$ corner:

$$f_{HP} = \frac{1}{2\pi R_9 C_5} = 234\,\text{Hz}. \tag{4}$$

The signal then passes through an LM358 unity-gain buffer U_1 (non-inverting input bias set by R_9). A passive low-pass section follows, with series $R_{10} = 100\,\Omega$ feeding a node loaded by $C_6 = 330\,\text{nF}$ in parallel with $R_{17} = 470\,\Omega$ to ground. R_{17} serves two roles simultaneously: as a DC divider it attenuates by $R_{17}/(R_{10} + R_{17}) = 0.825$, trimming the modulation depth from $84\,\%$ to the target $71\,\%$; and as part of the Thevenin resistance seen by C_6, it sets the upper corner

$$f_{LP} = \frac{1}{2\pi (R_{10} \parallel R_{17})\, C_6} = 5.85\,\text{kHz}. \tag{5}$$

The $234\,\text{Hz}$ HPF corner sits closest to the spec edge and proves the tighter of the two endpoints (Section VI).

B. Push-Pull Audio Buffer

The LM358 cannot supply the heavy current required by the transformer primary, whose impedance falls below $10\,\Omega$ at the low end of the audio band; a complementary emitter-follower pair (2N2222A Q_5 and 2N2907A Q_6) provides the necessary current gain. To prevent crossover distortion the bases must be biased $\sim 2V_{BE}$ apart. Since diodes are not permitted by the part list, we use a V_{BE} multiplier (Q_4, 2N2222A) with $R_{11}, R_{12} = 1\,\text{k}\Omega$ and chain resistors $R_{14}, R_{15} = 1\,\text{k}\Omega$ setting the quiescent point. Q_4's V_{CE} settles at $V_{BE}(1 + R_{12}/R_{11})$ and tracks temperature. $R_{13} = 27\,\Omega$ at the buffer output establishes a static DC reference for V_{buffer}.

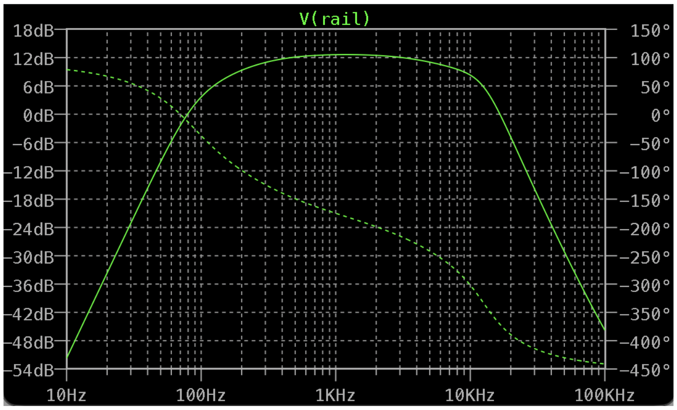

Fig. 2. Simulated audio-chain transfer function (AC sweep from 10 Hz to 50 kHz). Markers at the -3 dB points indicate the 234 Hz HPF and 5.85 kHz LPF corners, bracketing the required 250 Hz to 5000 Hz audio passband.

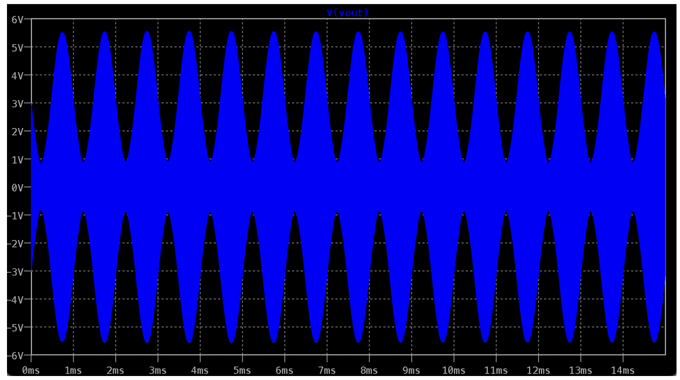

Fig. 3. Output voltage V_{out} across $R_L = 500\,\Omega$ over 14 ms with 1 kHz audio modulation. The 1 MHz carrier amplitude is modulated between approximately $0.93\,\mathrm{V_{pk}}$ and $5.5\,\mathrm{V_{pk}}$, consistent with the FFT-measured modulation depth of 71 %.

C. *Transformer and Rail Filtering*

The audio transformer is the Mouser 42TL001 ($L_p = 300\,\mathrm{mH}$, $n = 8.333$), connected in *reverse* so that the 4.32 mH winding ($1.1\,\Omega$ series resistance) is driven by the push-pull buffer and the 300 mH winding ($35\,\Omega$ series resistance) couples into the rail. This $8.33\times$ voltage step-up allows a modest buffer swing of $\pm 0.5\,\mathrm{V_{pk}}$ to produce the $\pm 4.2\,\mathrm{V_{pk}}$ of audio swing required for $m = 0.7$ on a 6 V rail.

The transformer secondary is placed in series with a 2.5 mH RFC (L_4) between V_{DD} and the Class D rail. The RFC presents high impedance to 1 MHz switching ripple while passing the audio signal essentially unattenuated, and a 100 nF bypass capacitor C_8 shunts residual RF content to ground.

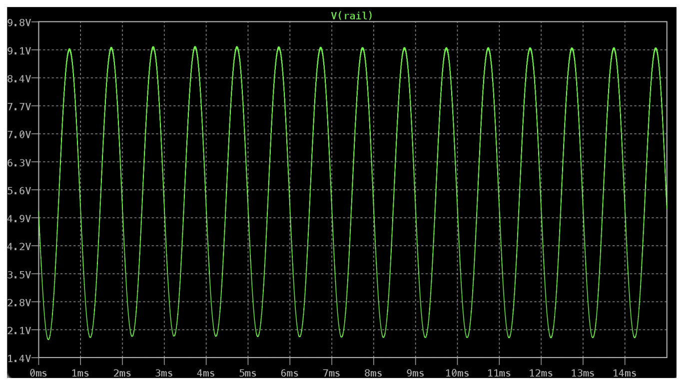

Fig. 4. Class D supply rail V_{rail} with 1 kHz audio. Sinusoidal envelope confirms low-distortion supply-rail modulation.

VI. Simulation Results

All simulations were performed in LTspice with the manufacturer SPICE models for the 2N2222A, 2N2907A, and LM358. The `.tran 0 20m 5m 20n` command was used with a 20 ns maximum timestep to resolve the 1 MHz carrier; the first 5 ms was discarded as startup transient. Per the spec, the SPICE `power` option is not used; all power quantities are computed from voltage and current via `.meas` commands.

A. *Modulated Output and Rail Waveform*

Fig. 3 shows the output voltage V_{out} across R_L with 1 kHz audio, displaying clean amplitude modulation of the 1 MHz carrier (envelope between $\sim 0.93\,\mathrm{V_{pk}}$ and $\sim 5.5\,\mathrm{V_{pk}}$). Fig. 4 shows the corresponding rail waveform: it swings between $V_{\text{rail,max}} = 9.19\,\mathrm{V}$ and $V_{\text{rail,min}} = 1.85\,\mathrm{V}$, giving a rail-based modulation depth

$$m_{\text{rail}} = \frac{V_{\max} - V_{\min}}{V_{\max} + V_{\min}} = 0.665. \tag{6}$$

B. *Spectrum and THD*

The FFT of V_{out} (Fig. 5) shows the carrier at 1 MHz flanked by the audio sidebands at $1\,\mathrm{MHz} \pm 1\,\mathrm{kHz}$. From the sideband-to-carrier amplitude ratio,

$$m_{\text{FFT}} = 2 \cdot \frac{A_{\text{SB}}}{A_C} = 0.71, \tag{7}$$

in good agreement with the rail-based estimate (the small discrepancy is due to slight envelope asymmetry, which the time-domain formula penalizes but the FFT does not). The second- and third-order audio sideband distortion products yield a combined THD of 3.4 %, well below the 10 % specification.

Carrier harmonics at 2 MHz and 3 MHz are attenuated by more than 28 dB relative to the unmodulated carrier, exceeding the 20 dB FCC mask by 8 dB of margin.

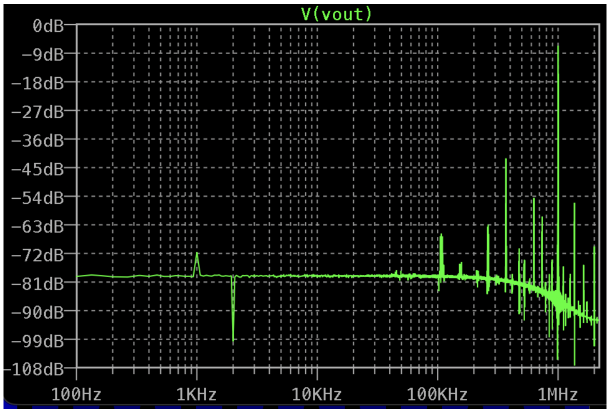

Fig. 5. FFT of V_{out} with $1\,\text{kHz}$ audio at $m = 0.7$. Carrier at $1\,\text{MHz}$ with audio sidebands at $\pm 1\,\text{kHz}$. Carrier harmonics at $2\,\text{MHz}$, $3\,\text{MHz}$ are $> 28\,\text{dB}$ below the fundamental.

C. Audio Passband Verification

The passband endpoints were verified at $250\,\text{Hz}$ and $5\,\text{kHz}$ audio. Both satisfy the $m \geq 0.707\,m_{1k} = 0.470$ ($-3\,\text{dB}$) criterion (Table I); Fig. 6 shows the corresponding rail waveforms with clean sinusoidal envelopes at both extremes. The $250\,\text{Hz}$ endpoint sits essentially at the threshold, consistent with the HPF corner at $234\,\text{Hz}$ being placed close to the spec edge for maximum in-band audio gain.

TABLE I
AUDIO PASSBAND VERIFICATION (RAIL-BASED MODULATION DEPTH)

Audio freq.	$V_{\text{rail,max}}$	$V_{\text{rail,min}}$	m_{rail}
250 Hz	8.31 V	2.75 V	0.503
1 kHz	9.19 V	1.85 V	0.665
5 kHz	8.68 V	2.37 V	0.572

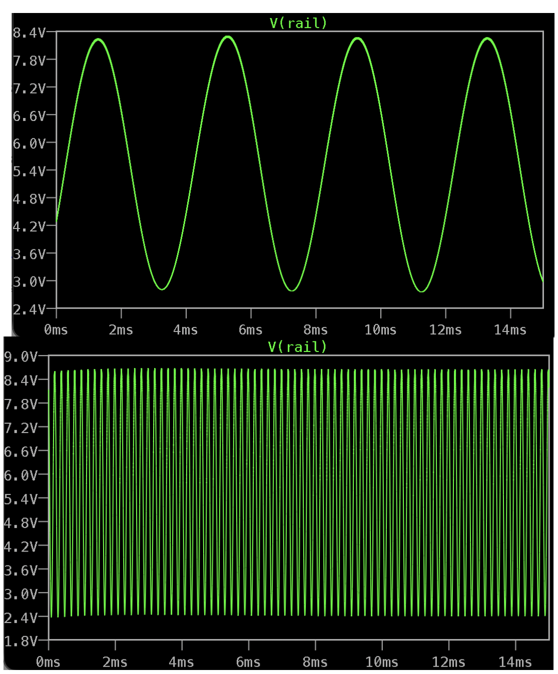

Fig. 6. Rail voltage at the audio passband endpoints. (Top) $250\,\text{Hz}$; (bottom) $5\,\text{kHz}$. Both envelopes are clean sinusoids.

D. Power Budget

Power results are summarised in Table II. The DC supply current averages $14.0\,\text{mA}$, giving $P_{DC} = V_{DD}\langle I_{L_3}\rangle = 84\,\text{mW}$. Output power into the load is $13.0\,\text{mW}$. $\langle V_{\text{rail}}\, I_{L_3}\rangle$ contains an audio cross-term from transformer pumping; per the specification the relevant figure is $V_{DD}\langle I_{L_3}\rangle$. An additional $1\,\text{k}\Omega$ resistor R_{18} at the rail node (a bench-test leftover) contributes $\sim 32\,\text{mW}$ of static load; even with this overhead the design clears the $100\,\text{mW}$ spec with $16\,\text{mW}$ of margin.

TABLE II
SIMULATED POWER BUDGET AT $1\,\text{kHz}$, $m = 0.7$

Quantity	Value
DC supply current $\langle I_{L_3}\rangle$	14.0 mA
DC dissipation $V_{DD}\langle I_{L_3}\rangle$	84 mW
Load power $V^2_{out,\text{rms}}/R_L$	13.0 mW
Specification limit	< 100 mW

VII. DISCUSSION

Two design decisions dominated the path to specification. First, supplying the RF buffer from the modulated rail rather than the static V_{DD} reduced THD from approximately $15\,\%$ to under $5\,\%$ with no other change; the mechanism is the elimination of a constant-amplitude pedestal in the base drive that would otherwise compress the modulation envelope at high m.

Second, the Class D base resistors $R_2 = R_3$ were swept from $1\,\text{k}\Omega$ to $2.2\,\text{k}\Omega$, dropping P_{DC} from $108\,\text{mW}$ to $84\,\text{mW}$—the largest single power improvement in the design. Lighter base loading reduces the rail current during conduction and lets the rail swing more fully, which in turn allows the same output power at lower DC.

Robustness to component tolerance was considered throughout. The HPF corner at $234\,\text{Hz}$ sits closest to its spec edge; a -5% shift in C_5 or R_9 would move it upward toward $250\,\text{Hz}$. We accepted this margin because the achieved $50.3\,\%$ modulation depth at $250\,\text{Hz}$ clears the $47.0\,\%$ threshold.

VIII. CONCLUSION

A Class D AM transmitter meeting all course and FCC Part 15 specifications was designed and verified in LTspice. The transmitter operates at $1\,\text{MHz}$ on a single $6\,\text{V}$ supply, dissipating $84\,\text{mW}$ DC in the final stage while delivering $13.0\,\text{mW}$ of audio-modulated RF into a $500\,\Omega$ load. Modulation depth, distortion, audio passband, and spurious emissions all satisfy the specification, with the FCC emission margin exceeding $8\,\text{dB}$ and THD a factor of three below the $10\,\%$ ceiling. Complete simulation results, additional figures, and design files are available in [4]

REFERENCES

[1] Y. Tsividis, *Operation and Modeling of the MOS Transistor*, 3rd ed. Oxford University Press, 2011.
[2] B. Razavi, *RF Microelectronics*, 2nd ed. Prentice Hall, 2012.
[3] H. L. Krauss, C. W. Bostian, and F. H. Raab, *Solid State Radio Engineering*. Wiley, 1980.
[4] A. Wang, “Class D AM Transmitter Design Repository,” 2026. [Online]. Available: https://github.com/Albert-Wang1010/comm-circuits-proj

Appendix A
Final Component Values

Table III lists the complete set of component values used in the final simulation. All resistors and capacitors are standard E12 values.

TABLE III
Final Component Values

Block	Component	Value
Audio chain	R_8 (source)	$50\,\Omega$
	R_{16} (shunt)	$4.7\,\mathrm{k\Omega}$
	C_5 HPF cap	$68\,\mathrm{nF}$
	R_9 HPF res	$10\,\mathrm{k\Omega}$
	U_1	LM358 (unity buffer)
	R_{10} LPF series	$100\,\Omega$
	R_{17} atten./LPF	$470\,\Omega$
	C_6 LPF cap	$330\,\mathrm{nF}$
	C_7 AC couple	$10\,\mathrm{\mu F}$
Push-pull buffer	Q_5 (NPN), Q_6 (PNP)	2N2222A / 2N2907A
	Q_4 (V_{BE} mult.)	2N2222A
	R_{11}, R_{12} (V_{BE} mult.)	$1\,\mathrm{k\Omega}$ each
	R_{14}, R_{15} bias chain	$1\,\mathrm{k\Omega}$ each
	R_{13} output bias	$27\,\Omega$
Trans- former / rail	T_1	42TL001 (reverse-wired)
	L_2 (driven)	$4.32\,\mathrm{mH}$, $R_{ser}=1.1\,\Omega$
	L_3 (rail)	$300\,\mathrm{mH}$, $R_{ser}=35\,\Omega$
	L_4 RFC	$2.5\,\mathrm{mH}$
	C_8 rail bypass	$100\,\mathrm{nF}$
	R_{18} rail shunt†	$1\,\mathrm{k\Omega}$
RF buffer	R_4 source res.	$50\,\Omega$
	C_3 AC couple	$100\,\mathrm{nF}$
	R_7 bias top	$7.5\,\mathrm{k\Omega}$
	R_5 bias bottom	$680\,\Omega$
	C_4 base bypass	$100\,\mathrm{pF}$
	Q_3	2N2222A
	R_6 collector (to rail)	$1\,\mathrm{k\Omega}$
Class D	Q_1 high-side PNP	2N2907A
	Q_2 low-side NPN	2N2222A
	R_2, R_3 base res	$2.2\,\mathrm{k\Omega}$
	C_2, C_9 speed-up	$330\,\mathrm{pF}$
	L_1 tank inductor	$820\,\mathrm{\mu H}$
	C_1 tank capacitor	$30\,\mathrm{pF}$
	R_1 load	$500\,\Omega$

† R_{18} was added for a bench test and retained in the final simulation; see Section VI.

Appendix B
Power Measurement Output

Fig. 7 reproduces the relevant section of the LTspice error-log output from the `.meas` commands at $1\,\mathrm{kHz}$ audio, $m = 0.7$.

```
i_l3_avg: AVG(i(l3))=-0.0140226 FROM 0 TO 0.015
p_dc: AVG(6*i(l3))=-0.0841355 FROM 0 TO 0.015
p_total_rail: AVG(v(rail)*i(l3))=-0.095379 FROM 0 TO 0.015
p_rfbuf: AVG(v(rail)*i(r6))=0.0256144 FROM 0 TO 0.015
p_classd: AVG(v(rail)*ie(q1))=-0.0169525 FROM 0 TO 0.015
p_load: AVG(v(vout)*i(r1))=0.0130176 FROM 0 TO 0.015
vout_rms: RMS(v(vout))=2.55124 FROM 0 TO 0.015
p_load_rms: vout_rms*vout_rms/500=0.0130176
v_rail_max: MAX(v(rail))=9.19234 FROM 0 TO 0.015
v_rail_min: MIN(v(rail))=1.8486 FROM 0 TO 0.015
v_rail_avg: AVG(v(rail))=5.51692 FROM 0 TO 0.015
v_sw_max: MAX(v(switchingnode))=9.36828 FROM 0 TO 0.015
v_sw_min: MIN(v(switchingnode))=-0.632456 FROM 0 TO 0.015
efficiency: p_load_rms/abs(p_dc)*100=15.4722
v_gap_peak: v_rail_max-v_sw_max=-0.175944
```

Fig. 7. LTspice `.meas` output (error-log excerpt) at $1\,\mathrm{kHz}$ audio, $m \approx 0.7$. Quantities used in the main body and power budget (Table II) are computed directly from these measurements.

Appendix C
Optimization History

The DC power figure converged across the following iterations (all measured at $1\,\mathrm{kHz}$ audio, $m \approx 0.7$):

1) Baseline (no speed-up, $R_{2,3} = 1\,\mathrm{k\Omega}$): $P_{DC} = 150\,\mathrm{mW}$.
2) Adding $330\,\mathrm{pF}$ speed-up caps: $P_{DC} = 133\,\mathrm{mW}$.
3) Increasing speed-up to $680\,\mathrm{pF}$: *worsened* to $140\,\mathrm{mW}$ (trough overshoot). Reverted.
4) Sweeping R_6: no net change in DC, only redistributes between p_{rfbuf} and p_{classd} via a cross-term.
5) Increasing R_2, R_3 from $1\,\mathrm{k\Omega}$ to $2.2\,\mathrm{k\Omega}$: $P_{DC} = 84\,\mathrm{mW}$. *Largest single improvement.*
6) Adding $R_{17} = 470\,\Omega$: trimmed m from $84\,\%$ to $71\,\%$, $P_{DC} = 84\,\mathrm{mW}$.

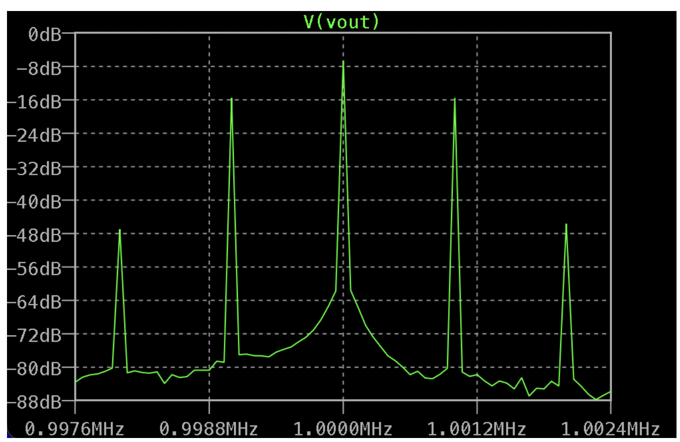

Fig. 8. FFT zoom near $1\,\mathrm{MHz}$ showing audio second- and third-harmonic distortion sidebands at $990\,\mathrm{kHz}$ and $985\,\mathrm{kHz}$ ($5\,\mathrm{kHz}$ audio). Both are more than $30\,\mathrm{dB}$ below the fundamental sideband.